

LEVERAGING AI FOR ENHANCED CRIME PREVENTION AND LAW ENFORCEMENT

(Master’s Thesis-Based)

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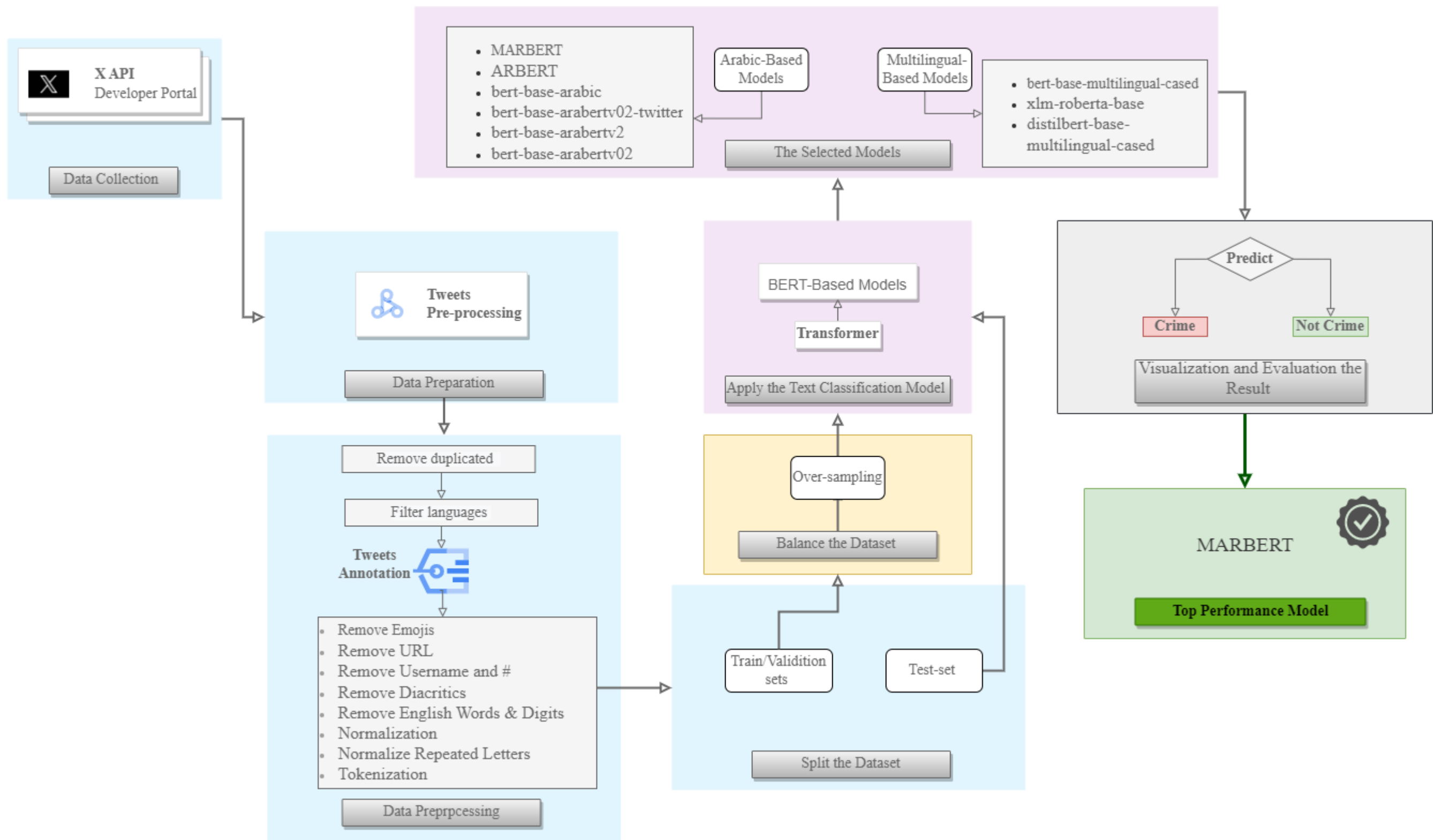
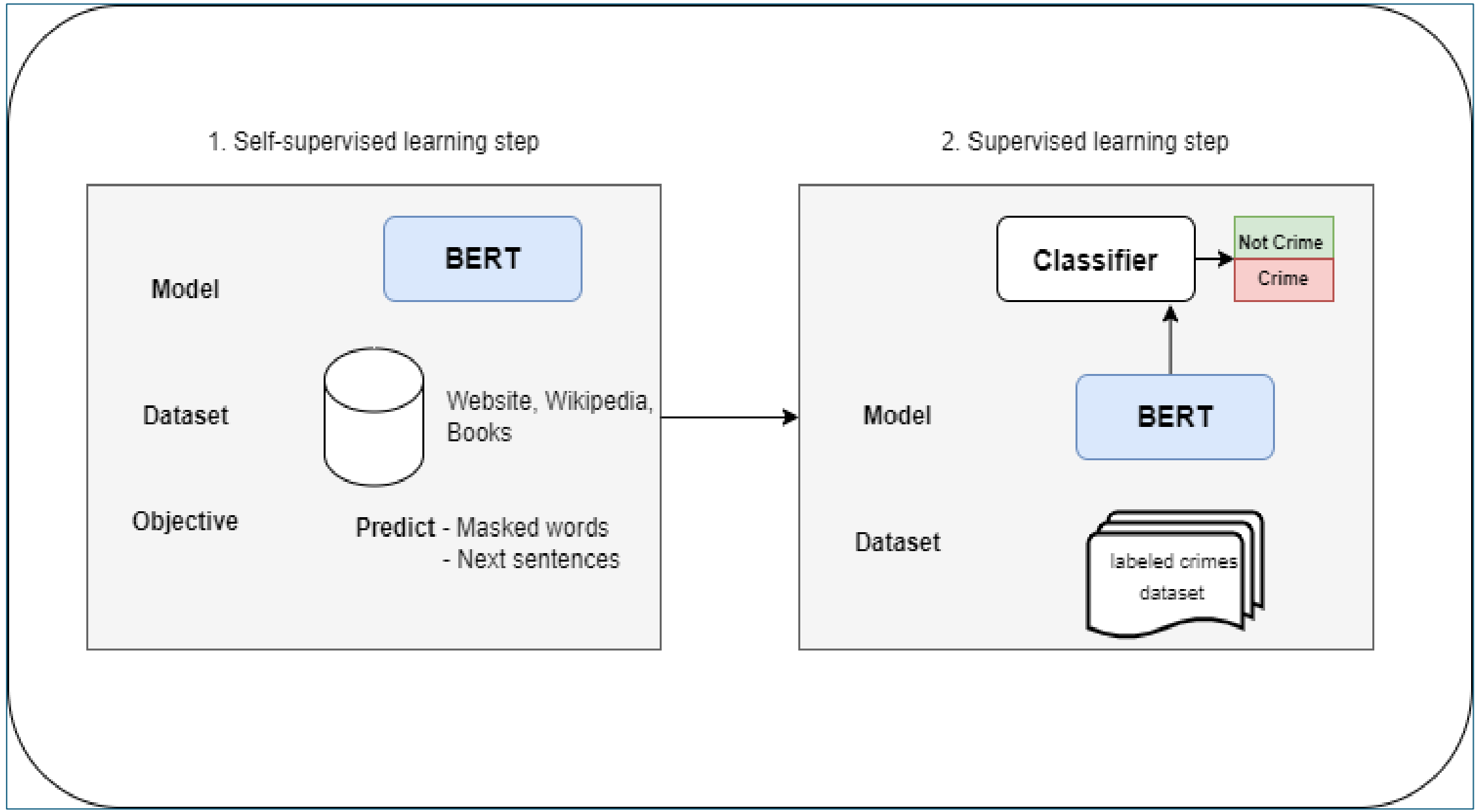
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1 INTRODUCTION

- Safety and security are fundamental pillars of social stability and national development.
- Social media platforms such as X generate large volumes of crime-related Arabic text daily.
- Manual analysis of this unstructured data is time-consuming and challenging for law enforcement agencies.
- Recent advances in LLMs, especially BERT models, have shown strong performance in text classification tasks.
- However, Arabic crime classification remains underexplored, and publicly available datasets are limited.
- This study fine-tunes Arabic and multilingual BERT-based models for Arabic crime classification to support crime prevention and law enforcement efforts.

2 METHODOLOGY

The proposed framework consists of several stages for Arabic crime tweet classification:

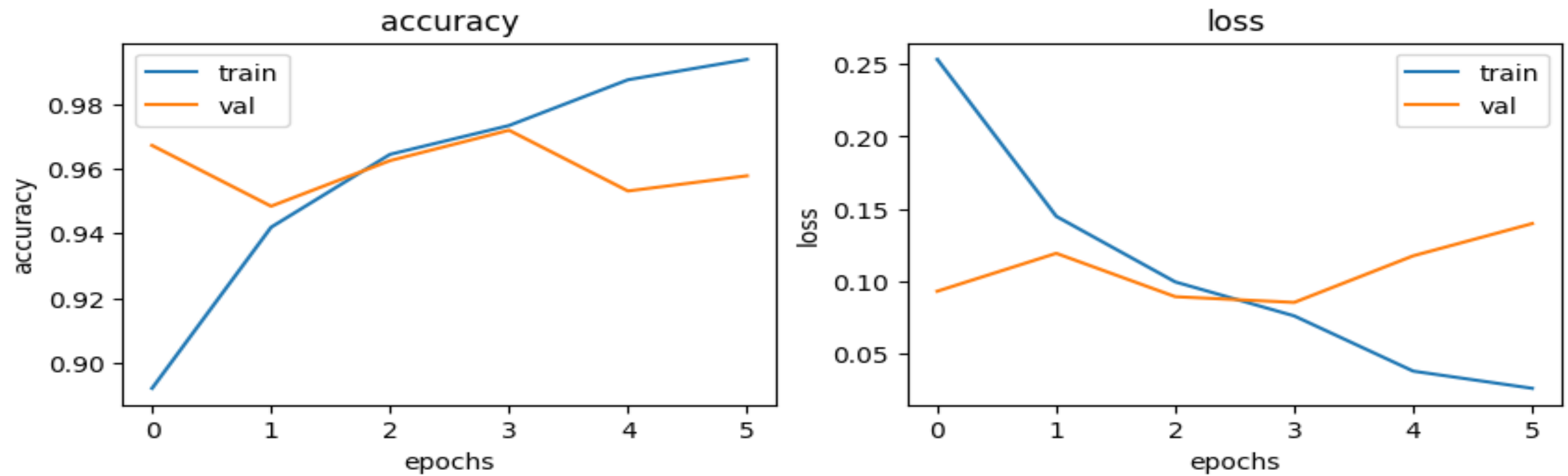
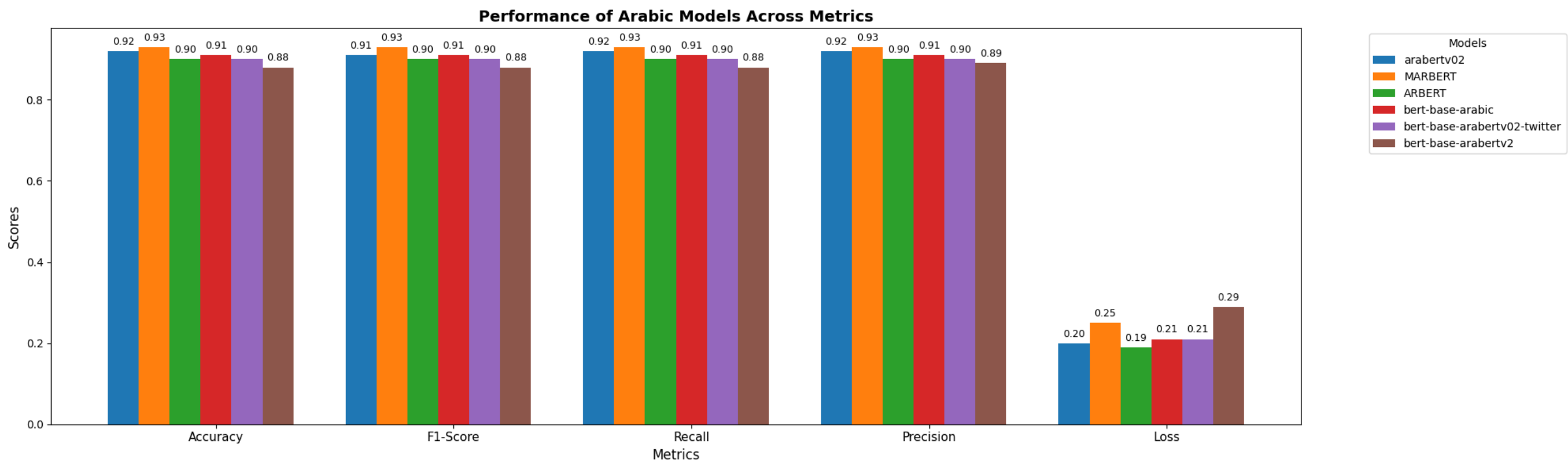


3 EXPERIMENT

Experiments were conducted to compare Arabic and multilingual BERT-based models for crime tweet classification.

Key Findings

- ✓ Arabic-specific models significantly outperformed multilingual models.
- ✓ MARBERT achieved the best overall performance.
- ✓ Multilingual BERT models showed lower effectiveness on Arabic crime-related text.



Additional Findings

- ✓ Arabic pretrained models better captured dialectal and contextual Arabic language patterns.
- ✓ Pre-processing and balancing techniques significantly improved classification performance.
- ✓ The proposed framework outperformed several previous crime classification approaches.

Model /Study	Accuracy (%)	F1-score (%)	Recall (%)	Precision (%)
AraBERT/ (Mubarak et al., 2020)	-	83	82	85
MARBERT/ (Shapiro et al., 2022)	86	84	84	84
AraBERT/(El-Alami et al., 2022)	91	93	-	-
MARBERT/ ours	93	93	93	93
AraBERT/ ours	92	91	92	92

4 CONCLUSION

This study fine-tunes Arabic and multilingual BERT models for Arabic crime classification. The results show that Arabic-specific models outperform multilingual models, with MARBERT achieving the best performance, reaching 93% accuracy and F1-score. The proposed framework supports crime detection and contributes to crime prevention efforts for law enforcement agencies.

Future work includes:

- ❖ Expanding the dataset
- ❖ Multi-class crime classification
- ❖ Real-time crime monitoring systems